

WAZUH Home Lab -SIEM and File Integrity Monitoring

1. Overview

- Wazuh is open-source security platform which have Log Analysis, File Integrity Monitoring, and Vulnerability detection.

2. Pre -Setup

- Wazuh Manager Ubuntu Server latest version (VirtualBox)
- Wazuh Agent Windows (host machine) and Administrative access

3. Installing the Wazuh Manager (Ubuntu)

1. Turn on your Ubuntu VirtualBox Server
2. Open the terminal window and type "**curl -s https://packages.wazuh.com/key/GPG-KEY-WAZUH | sudo gpg --dearmor -o /usr/share/keyrings/wazuh-archive-keyring.gpg**" - This will add the GPG key to verify Wazuh packages
3. In the same terminal window type "**curl -sO https://packages.wazuh.com/4.12/wazuh-install.sh && sudo bash ./wazuh-install.sh -a -l**"

4. Accessing the Wazuh Dashboard

1. Once installation is complete check the ip address of the Ubuntu by type "**ifconfig**" in the terminal window.
2. On Ubuntu open up the browser and type "**https://<ubuntu-vm-ip>**"
3. Accept the browser security warning and login using credential

5. Installing the Wazuh Agent (Windows Host)

1. Turn on your Windows System
2. Download the latest Wazuh agent MSI installer
3. Install the MSI package on your Windows system

6. Registering the Agent with the Manager

1. Run the agent management and type "**sudo /var/ossec/bin/manage_agents**"
2. Select A to add an agent
3. Assign a name (for example master system)
4. Leave IP address blank
5. After creation, select E to extract the key.
6. Copy the key output.c
7. Open Wazuh Agent Manager GUI from the Start Menu.
8. Paste the copied key into the appropriate field.
9. Save and apply the key.
10. Add the manager's IP address (IP address of your Ubuntu manager).
11. Restart the agent service.
12. Go to WAZUH dashboard and see the agent onboarded.

7. File Integrity Monitoring on Windows

1. Open the following configuration file:
2. C:\Program Files (x86)\ossec-agent\ossec.conf
3. Add the following entry inside the Directory block:
4. **<directories realtime="yes"> C:\Users\abc\Test</directories>**
5. Restart the Wazuh agent service to apply the configuration.

8. Verifying Setup

1. Open the Wazuh Dashboard
2. Navigate to Agents check if Windows agent is listed and status is Active.
3. Go to the Integrity Monitoring section.
4. Perform actions (create/modify/delete files) in the monitored folder.
5. Confirm that alerts appear in the dashboard.

Tutorial document is complete